

✓
PASSED

ALL 7 STREAMING QoE SERVICE LEVEL AGREEMENTS (SLAs) MET

OVERALL MOS: 4.28 / 5.0 (EXCELLENT (Tier 1 OTT))

EXECUTIVE SUMMARY & BUSINESS IMPACT

This automated QoE audit evaluated player playback continuity, startup latency, and transport resilience across **MPEG-DASH** and **Apple HLS** under extreme stress conditions (CDP network drenching, 6-second total network blackout, and rapid multi-phase bandwidth turbulence). With an initial startup latency of **209 ms** and a resilient ABR Stability Index of **100%**, the player engine avoids viewer drop-off during startup and prevents churn-inducing oscillation storms.

SERVICE LEVEL AGREEMENT (SLA) VERIFICATION MATRIX

BENCHMARK SLA	TARGET OBJECTIVE	OBSERVED TELEMETRY	SLA STATUS	BUSINESS IMPACT
1. Startup Latency (TTFF)	< 3,500 ms	209 ms	PASS	Minimizes viewer abandonment during playhead startup.
2. Seek Recovery Resilience	0 fatal pipeline freezes	Recovered (6 rapid hops)	PASS	Maintains MediaSource pipeline integrity under erratic scrubbing.
3. ABR Downswitch Adaptation	Sustain playback < 350 kbps	14932 → 254 kbps	PASS	Graceful resolution downgrade prevents catastrophic video stall.
4. Outage Recovery (RRT)	Re-buffer recovery < 6,000 ms	6 ms (Buffer: 10.9s)	PASS	Instantaneous recovery after subway/tunnel network dropouts.
5. ABR Stability Index	≥ 70 % stability score	100 %	PASS	Suppresses ABR flip-flop oscillation storms during cellular jitter.
6. Protocol Parity (DASH vs HLS)	Both TTFF < 10s & MOS ≥ 1.0	DASH: 267ms HLS: 1143ms	PASS	Validates cross-format delivery parity for Web & Apple ecosystems.
7. Multi-Audio & Subs SLA	Zero-deadlock language switch	5 Audio / 4 Subs tracks	PASS	Ensures seamless multi-language internationalization without A/V drift.

⚡ Protocol Face-Off (400 kbps Throttle)

SLA MET

DASH Startup (TTFF): 254 ms (Winner 🚀)

HLS Startup (TTFF): 1251 ms

DASH Throttled MOS: 1.27 / 5.0

HLS Throttled MOS: 1.23 / 5.0 (Winner 💎)

DASH vs HLS Parity: Both Streams Resilient

🌀 Transport Chaos & Chaos Engineering

STABLE

Turbulence Profile: 5-Phase Rollercoaster (12M ↔ 250k)

Adaptations Triggered: 6 clean variant shifts

Rebuffering Recovery: Immediate (< 10 ms)

Frame Drop Ratio: 0% (0 dropped)

Active Variant Codec: 1280x720 (H.264 / AVC1)